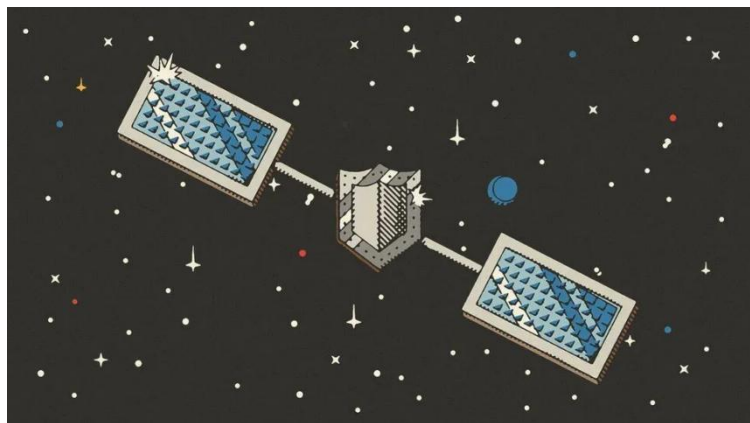


考研精读 | Economist



Taking the hit

迎接冲击

Satellites get a new type of **armour**

人造卫星的新盔甲

难度评级：★★★★☆

【 Para.1 】

Hurting around Earth at roughly seven kilometres a second, even tiny pieces of **debris pack quite a punch**. Such manmade junk **poses** a far greater threat to satellites circling in low **orbits** than natural space rocks do—and the risks are rising. According to the European Space Agency, the amount of rubbish in orbit has more than doubled since 2017.

若以每秒约 7 公里的速度绕地球飞驰，即便是微小的碎片也具有相当大的破坏力。与天然太空岩石相比，这些人造垃圾对在低轨道上运行的卫星构成了更大的威胁——而且风险还在不断上升。据欧洲航天局称，自 2017 年以来，轨道上的垃圾数量已翻了一番多。

【 Para.2 】

September 12TH-18TH | Science & technology

文章导读：

近地轨道太空垃圾数量持续增长，高速碎片严重威胁卫星安全，不少卫星故障疑似源于碎片撞击。传统惠普尔防护盾笨重，铝制防护能力有限。目前多款新型轻量化防护方案相继问世，包括复合板材、凯夫拉陶瓷材料、3D 打印多孔护盾与泡沫铝。配套冲击传感装置可采集轨道撞击数据，助力卫星防护设计更精准。

armour n. /'ɑ:mə(r)/ [U] special metal clothing that soldiers wore in the past to protect their bodies while fighting 盔甲；甲冑【考研英语词汇】

hurtle v. /'hɜ:tl/ to move very fast in a particular direction (向某个方向) 飞驰，猛冲

A runaway car came **hurtling** towards us. 一辆失控的汽车朝我们飞驰而来。

debris n. /'debri:/ [U] (formal) pieces of material that are not wanted and rubbish/garbage that are left somewhere 残渣；垃圾；废弃物【考研英语词汇】

pack a (powerful, real, etc.) punch idiom to have a powerful effect on sb 产生巨大影响；十分有效力【考研英语词汇】 The advertising campaign packs quite a punch. 这次广告造势产生了相当大的影响。

pose v. to create a threat, problem, etc. that has to be dealt with 造成（威胁、问题等）；引起；产生【考研英语词汇】 to pose a threat/challenge/danger/risk 构成威胁/挑战/危险/风险

orbit n. /'ɔ:bit/ a curved path followed by a planet or an object as it moves around another planet, star, moon, etc. (天体等运行的) 轨道【考研英语词汇】

collision n. /kə'liʒn/ an accident in which two vehicles or people crash into each other 碰撞（或相撞）事故【考研英语词汇】

adhesive a. /əd'hi:sv/ that can stick to sth 黏合的；黏附的；有附着力的【考研英语词汇】

strip n. a long narrow piece of paper, metal, cloth, etc. (纸、金属、织物等) 条，带【考研英语词汇】

outlying a. Outlying places are far away from the main cities of a country. 边远的

bumper n. /'bʌmpə(r)/ a bar fixed to the front and back of a car, etc. to reduce the effect if it hits anything (汽车头尾的) 保险杠

aluminium n. /æljə'mɪniəm/ [U] a chemical element. Aluminium is a light, silver-grey metal used for making pans, etc. 铝【考研英语词汇】

spray n. /sprei/ very small drops of a liquid that are sent through the air, for example by the wind 浪花；水花；飞沫【考研英语词汇】

fragment n. /'frægmənt/ a small part of sth that has broken off or comes from sth larger 碎片；片段【考研英语词汇】

Roughly one small satellite in five stops working within a year of launch, estimates a study published in 2021, and in three-fifths of these cases operators do not know what happened. The smart money is on quite a few such failures being caused by **collisions** with space debris. To provide more data Odin Space, a firm based in Santa Ana, California, has begun selling a system that uses vibration-sensing **adhesive strips** to detect and assess impacts. The results should help settle bets on the best shielding for a given orbit.

2021 年发表的一项研究估计，大约五分之一的小型卫星在发射后一年内就会停止工作，而在其中五分之三的案例中，运营商甚至并不知道具体情况。有充分理由相信，相当一部分此类故障是由与太空垃圾的碰撞引起的。为了提供更多数据，总部位于加利福尼亚州圣安娜的奥丁太空公司（Odin Space）已开始销售一种系统，该系统利用振动感应粘合条来检测和评估撞击。研究结果应有助于确定针对特定轨道最有效的防护方案。

【 Para.3 】

One long-standing approach to protecting satellites is a Whipple shield—a two-stage affair which places an **outlying bumper** a few centimetres above a main shield, both of which are typically made of **aluminium**. The bumper breaks an incoming object into a **spray** of smaller, less energetic **fragments** that the main shield can then stop more easily. But this arrangement is **bulky** and heavy. So most satellites simply take their chances behind aluminium

bulky a. (of a thing东西) large and difficult to move or carry 庞大的；笨重的

stiffen v. ~ sth (with sth) to make sth, such as cloth, firm and unable to bend 使硬挺

honeycomb n. /'hʌnikəʊm/ a structure of cells with six sides, made by bees for holding their honey and their eggs 蜂巢

composite a. /'kɒmpəzɪt/ [only before noun] made of different parts or materials 合成的；混成的；复合的【考研英语大纲词汇】

cagey a. /'keɪdʒi/ ~ (about sth) (informal) not wanting to give sb information 守口如瓶的；讳莫如深的

SYN evasive, secretive

Tony is very **cagey** about his family. 托尼对自己家的事讳莫如深。

projectile n. /prə'dʒektail/ any object that is thrown as a weapon (作为武器的) 发射物；导弹

shrapnel n. /'ʃræpnəl/ [U] small pieces of metal that are thrown up and away from an exploding bomb 飞溅的弹片

metallic a. /mə'tælɪk/ [usually before noun] made of or containing metal 金属制的；含金属的

vaporise v. If a liquid or solid vaporizes or if you vaporize it, it changes into vapor or gas. (使)蒸发

The benzene **vaporized** and formed a huge cloud of gas. 苯蒸发形成了一大团气体。

-proof a. -proof combines with nouns and verbs to form adjectives indicating that something cannot be damaged or badly affected by the thing or action mentioned. 抗...的 (与名词或动词结合构成形容词)【考研英语大纲词汇】

employ v. If you employ certain methods, materials, or expressions, you use them. 使用【考研英语大纲词汇】

The group will **employ** a mix of tactics to achieve its aim. 该团体将运用混合策略来实现其目标。

experimental a. Something that is experimental is new or uses new ideas or methods, and might be modified later if it is unsuccessful. 试验性的【考研英语大纲词汇】

void n. If you describe a situation or a feeling as a void, you mean that it seems empty because there is nothing interesting or worthwhile about it. 空白；空虚感【考研英语大纲词汇】

subject v. ~ sb/sth to sth [often passive] to make sb/sth experience, suffer or be affected by sth, usually sth unpleasant 使经受；使遭受【考研英语大纲词汇】

The city was **subjected** to heavy bombing. 那座城市遭受猛烈轰炸。

successive a. /sək'sesɪv/ [only before noun] following immediately one after the other 连续的；接连的；相继的【考研英语大纲词汇】

means n. ~ (of doing sth/of sth) an action, an object or a system by which a result is achieved; a way of achieving or doing sth 方式；方法；途径【考研英语大纲词汇】

ceramic a. /sə'remɪk/ a pot or other object made of clay that has been made permanently hard by heat 陶瓷制品；陶瓷器

panels **stiffened** with a **honeycomb** also made of that metal.

These offer only limited protection from debris.

保护卫星的一种传统方法是采用惠普尔（Whipple）防护罩——这是一种两级结构，在外层防护罩上方几厘米处设置一个外围缓冲器，两者通常均由铝制成。缓冲器会将将来袭物体击碎成许多较小、动能较低的碎片，主防护罩随后就能更轻松地拦截这些碎片。但这种结构笨重且体积庞大。因此大多数卫星只能寄希望于铝制蜂窝结构加固的铝板来抵御撞击。而这些措施对太空碎片的防护作用又十分有限。

【 Para.4 】

Yet the range of options for protecting satellites is increasing. One of the alternatives on offer is **composite** tiles. Atomic-6, a firm near Atlanta, is developing these under the brand name Space Armor. It is **cagey** about the materials involved, but the resulting tiles are hand-size, 2.5cm thick and weigh about as much as an iPhone. In tests, they successfully stop aluminium **projectiles** the size of a pea that have been accelerated by a gas gun to 7.2km a second. Moreover, unlike Whipple shields, they throw off no **shrapnel** that could become someone else's problem, says Trevor Smith, Atomic-6's boss. Instead, incoming metal objects (and much space junk is **metallic**) are **vaporised** and absorbed.

然而，保护卫星的方案正日益丰富。其中一种替代方案是复合材料防护板。位于亚特兰大附近的一家企业 Atomic-6 正在以“Space Armor”为品牌名称开发此类产品。该公司对所用材料守口如瓶，但成品防护板大

porous a. /'pɔːrəs/ having many small holes that allow water or air to pass through slowly 多孔的；透水的；透气的

cavity n. /'kævəti/ a hole or empty space inside sth solid 洞；孔；窟窿；腔

scatter v. /'skætə(r)/ to move or to make people or animals move very quickly in different directions 散开；四散；使分散；驱散【考研英语大纲词汇】

SYN disperse

At the first gunshot, the crowd **scattered**. 枪声一响，人群便逃散了。

repertoire n. /'repətwaː(r)/ all the things that a person is able to do (某人的) 全部才能，全部本领【考研英语大纲词汇】

hinge v. ~on/upon sth (of an action, a result, etc. 行动、结果等) to depend on sth completely 有赖于；取决于【考研英语大纲词汇】

Everything **hinges** on the outcome of these talks. 一切都取决于这些会谈的结果。

fleck n. /flek/ [usually pl.] a very small area of a particular colour 色斑；斑点【考研英语大纲词汇】

prevail v. If a situation, attitude, or custom prevails in a particular place at a particular time, it is normal or most common in that place at that time. 盛行；普遍存在【考研英语大纲词汇】

A similar situation **prevails** in Canada. 同样的情况在加拿大也普遍存在。

gird v. If you gird yourself for a battle or contest, you prepare yourself for it. 装备

With audiences in the U.S. falling for the first time in a generation, Hollywood is **girding** itself for recession.

由于美国观众数量在这一代第一次出现下降，好莱坞正在为衰退做准备。

小如手掌，厚 2.5 厘米，重量约等于一部 iPhone。在测试中，这些瓷砖成功拦截了由气体枪加速至每秒 7.2 公里、豌豆大小的铝制弹丸。此外，Atomic-6 公司负责人特雷弗·史密斯表示，与惠普尔防护罩不同，它们不会产生可能给他人带来麻烦的弹片。相反，飞来的金属物体（而许多太空垃圾都是金属的）会被汽化并吸收。

【 Para.5 】

Other types of shielding are also advancing. One combines Kevlar, a fibre used to make bullet-**proof** jackets (and which also helps protect the International Space Station), with Nextel, a woven ceramic. This promises light shields for smaller, less expensive spacecraft—weight being a crucial consideration for launch costs.

其他类型的防护盾也在不断进步。其中一种将凯夫拉尔（一种用于制造防弹背心、同时也用于保护国际空间站的纤维）与 Nextel（一种编织陶瓷）相结合。这有望为更小、更经济的航天器提供轻质防护盾——重量是影响发射成本的关键因素。

【 Para.6 】

Another approach, from a team at the University of Padua, in Italy, **employs** 3D printing to up the ante. Lorenzo Olivieri and his colleagues have made **experimental** shields using aluminium, Kevlar and resin reinforced with carbon fibres. Printing such shields layer by layer allows internal cavities to be introduced. **These voids**, says Dr Olivieri, act

►今日句子分析

Para 6. These voids, says Dr Olivieri, act as small Whipple shields, subjecting incoming projectiles to successive shocks that serve to break them into ever-tinier fragments. 奥利维耶里博士表示，这些空隙起到了微型惠普尔防护罩的作用，使来袭的弹丸受到连续冲击，从而将其击碎成越来越小的碎片。

宾语从句		主句		
These voids,		says Dr Olivieri,	act	as small Whipple shields,
主语		谓语	主语	谓语
subjecting		incoming projectiles	to successive shocks	
伴随状语		subject 的宾语		状语
定语从句				
that	serve	to break them into ever-tinier fragments.		

从句主语 谓语 目的状语

主句：

谓语-says；主语-Dr Olivieri；

宾语从句：

主语 -These voids；谓语 -act；补语 -as small Whipple shields；

伴随状语 -subjecting incoming projectiles to successive shocks；

定语从句，修饰shocks：

从句主语-that；谓语-serve；目的状语-to break them into ever-tinier fragments（break sth into sth “把.....打碎/分解成.....”，后面的 ever-tinier 强调“越来越小的”，ever-+比较级/现在分词/形容词等，突出“一直处于.....的状态或者趋势”，可以用来作前置定语，修饰名词，类似：...the ever-increasing traffic on our roads. ...道路上不断增长的交通量。）。

as small Whipple shields, **subjecting** incoming projectiles to **successive** shocks that serve to break them into ever-tinier fragments.

另一种方法来自意大利帕多瓦大学的一个团队，他们利用 3D 打印技术进一步提升了防护水平。洛伦佐·奥利维耶里（Lorenzo Olivieri）及其同事利用铝、凯夫拉尔纤维以及碳纤维增强树脂制造了实验性防护罩。通过逐层打印这种防护罩，可以在内部形成空腔。奥利维耶里博士表示，这些空隙起到了微型惠普尔防护罩的作用，使来袭的弹丸受到连续冲击，从而将其击碎成越来越小的碎片。

【 Para.7 】

A still-further **means** to deliver Whipple-mimicking repeated shocks is with aluminium foam. This is made by packing a **ceramic** material into a **porous** plastic structure, then burning away the plastic to leave a network of **cavities** into which molten aluminium is poured. Knocking the ceramic out leaves behind a metal matrix that has voids up to 5mm in diameter **scattered** throughout its interior. Testing by NASA, America's space agency, found that a shield incorporating a layer of this foam a little over 6mm thick could provide the same protection as a conventional Whipple shield of twice the weight.

另一种模拟惠普尔效应、产生连续冲击的方法是使用铝泡沫。其制备过程是将陶瓷材料填充到多孔塑料结构中，随后烧掉塑料，留下一个空腔网络，再将熔融铝浇铸其中。敲出陶瓷后，便会留下一个金属基体，

➤ 写作积累

Para.5 This promises light shields for smaller, less expensive spacecraft—weight being a crucial consideration for launch costs.

这里的 promise 意为“有望带来、预示着、有可能提供”，类似于 can offer.....，文中这里因而指“这有望为更小、更便宜的航天器带来轻量化的防护罩”。而后半部分跟的 weight being a crucial consideration for launch costs 则是一个独立主格结构，具体为：名词/代词 + 现在分词短语/过去分词短语/介词短语/形容词.....。

文中的这一部分作状语，补充说明为什么防护罩也会对发射有影响。相当于：..., because weight is a crucial consideration for launch costs. 即“轻量防护罩之所以重要，是因为重量直接关系到发射成本”。而相对于 because，独立主格结构可以省去 because/since/and 等连接词，更紧凑，适合学术、科技、新闻写作。

Para.8 That will make girding for orbit a bit less of a shot in the dark.

be of + 抽象名词可以表示“具有某种性质/身份”，例如：be of importance = be important。而加了 more/less 会更突出前后或者与其他事物的比较

be more of + 名词 = 更像是.....；更大程度上是.....

be less of + 名词 = 不太像.....；较小程度上是.....

例如：He is more of a teacher than a researcher. 他更像老师，而不是研究者。

文中还有一处 shot in the dark 值得参考。这个短语是一个习语，从字面也可以猜出它意为“盲目猜测、瞎猜、没有把握的尝试”，例：

The figure he came up with was really just a shot in the dark. 他得出的数字实际上只是瞎猜而已。

其内部散布着直径达 5 毫米的空隙。美国国家航空航天局（NASA）的测试发现，采用一层厚度略超过 6 毫米的这种泡沫制成的防护盾，能提供与重量为其两倍的常规惠普尔防护层相同的防护效果。

【 Para.8 】

However promising this **repertoire** of shielding types is, though, much will **hinge** on what they encounter. Projectiles range from chunks of metal and pieces of plastic to rocks and **flecks** of paint, and the mix varies with the orbit, notes Rannveig Marie Faergestad, a shielding designer at Thales Alenia Space, a Franco-Italian spacecraft-maker. Better

knowledge of what **prevails** in a given zone would therefore help, and Odin Space's impact sensors should provide it. That will make **girding** for orbit a bit less of a shot in the dark.

尽管这些防护类型前景可期，但其效果很大程度上取决于实际遭遇的物体。法意合资航天器制造商泰雷兹阿莱尼亚航天公司的防护设计师兰维格·玛丽·费尔格斯塔德指出，撞击物种类繁多，从金属块、塑料碎片到岩石和油漆碎屑不等，且具体组合会因轨道不同而变化。因此，更深入地了解特定区域内主要存在的物体将大有裨益，而奥丁太空公司的撞击传感器应能提供这些信息。这将为为进入轨道做准备的工作不再那么像是在“瞎摸索”。